

Problem Set 8 – Packaging & Refactoring the Local AI Text Utility

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Package: vsearch.py

```
1  """vsearch: text search tools for AI pre-processing.
2
3  Packaged with setup.py (py_modules=['vsearch']), built via
4  'python3 -m build --sdist', and installed with pip so it can be
5  imported from anywhere, not just this folder. Passes pycodestyle
6  clean (see linting/ for the before/after run).
7
8  Author: Kshitiz
9  """
10
11
12 def search4letters(phrase: str, letters: str = 'ATCG') -> set:
13     """Return the set of target letters found in a supplied phrase."""
14     return set(letters).intersection(set(phrase))
```

Package: setup.py

```
1  from setuptools import setup
2
3  setup(
4      name='vsearch',
5      version='1.0',
6      description='Text search tools for AI pre-processing',
7      author='Kshitiz',
8      author_email='student@school.edu',
9      url='localhost',
10     py_modules=['vsearch'],
11 )
```

Trace Log

```
1  --- Task 1: Build, install, verify import ---
2  $ python3 -m build --sdist
3  ...
4  Successfully built vsearch-1.0.tar.gz
5
6  $ python3 -m pip install dist/vsearch-1.0.tar.gz
7  Successfully installed vsearch-1.0
8
9  $ cd /tmp
10 $ python3
11 >>> import vsearch
12 >>> vsearch.search4letters('TGGACC', 'GC')
13 {'G', 'C'}
14 >>> vsearch.__file__
15 '/usr/local/lib/python3.12/dist-packages/vsearch.py'
16
```

```
17 --- Task 2: Linting, before fixes ---
18 $ python3 -m pycodestyle vsearch_flawed.py
19 vsearch_flawed.py:1:26: E231 missing whitespace after ':'
20 vsearch_flawed.py:2:71: W291 trailing whitespace
21 vsearch_flawed.py:4:1: E302 expected 2 blank lines, found 0
22
23 --- Task 2: Linting, after fixes ---
24 $ python3 -m pycodestyle vsearch_fixed.py; echo "Exit code: $?"
25 Exit code: 0
26
27 --- Edge case: 4th flaw (line too long) ---
28 $ python3 -m pycodestyle vsearch_e501.py
29 vsearch_e501.py:2:80: E501 line too long (111 > 79 characters)
```

Submission Note

import vsearch succeeded from /tmp, well outside text_utility_pkg, and vsearch.__file__ pointed to the installed copy under dist-packages, not the original script. Testing the import from inside text_utility_pkg wouldn't have proven anything, because Python checks the current directory before site-packages – it would find vsearch.py sitting right there on disk and import that directly, which works whether or not the package was ever actually built or installed. Running from a completely different folder is the only way to confirm pip install really put a usable copy somewhere Python's import system finds on its own.

Linting caught all three injected flaws by their expected codes (E231, W291, E302) at the exact line/column reported, and the fixed file passed silently with exit code 0. A fourth injected flaw (a line over 79 characters) was caught as E501, a code not seen in the first three fixes.